MECH 325
Flexible Drive Elements



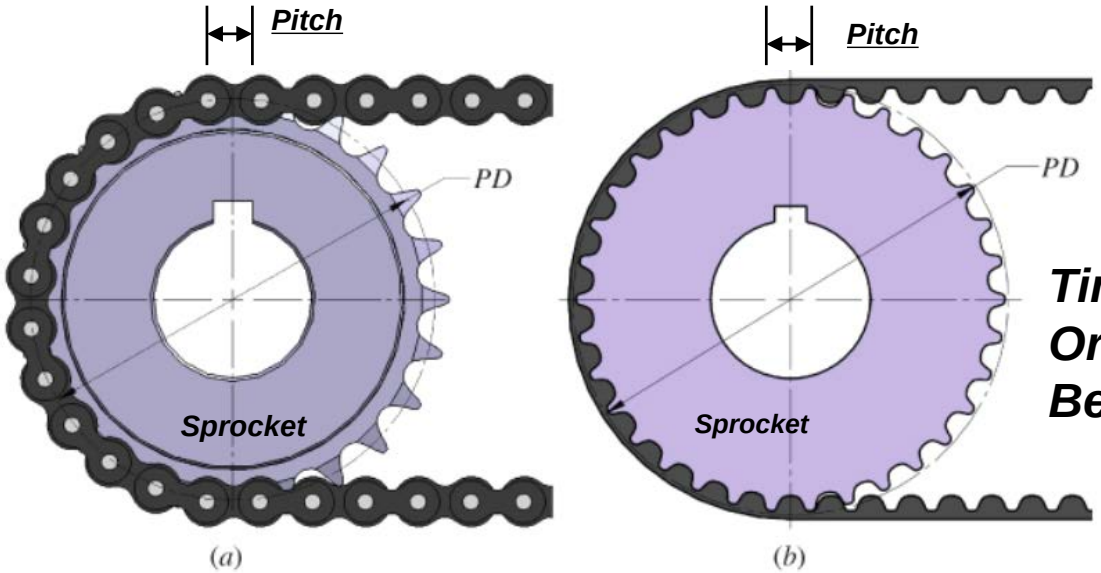
Objectives

By the end of this section, you should be able to:

- Describe the characteristics and uses of:
  - Flat belts
  - V-belts
  - Timing belts
  - Roller chains
- Analyze flexible drives in terms of power, speed, forces, and life
- Select and size an appropriate flexible drive for a given application

Terminology

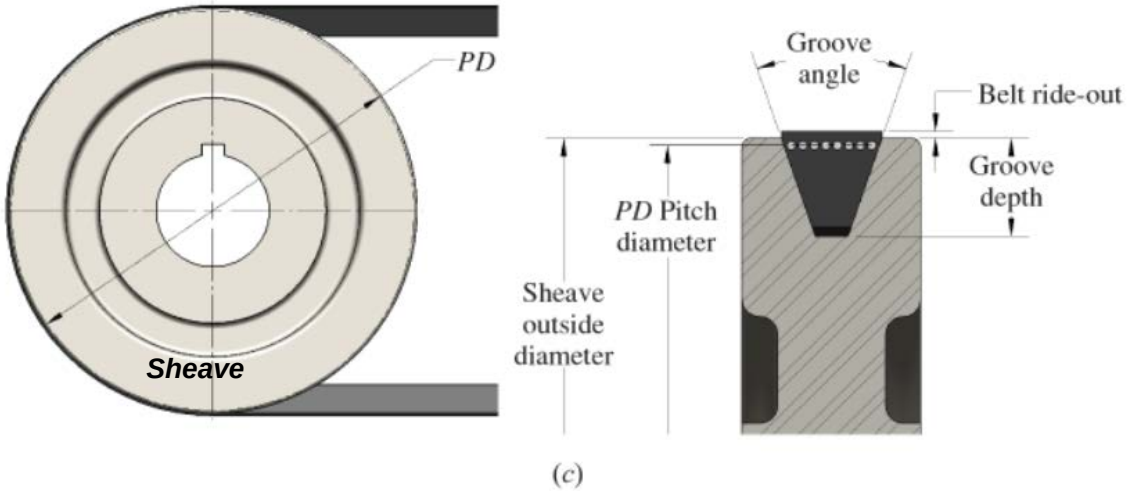
Roller Chain



Timing, Toothed Or Synchronous Belts

Vee Belt

"V-Belts"



Flexible Drive Overview Team Exercise

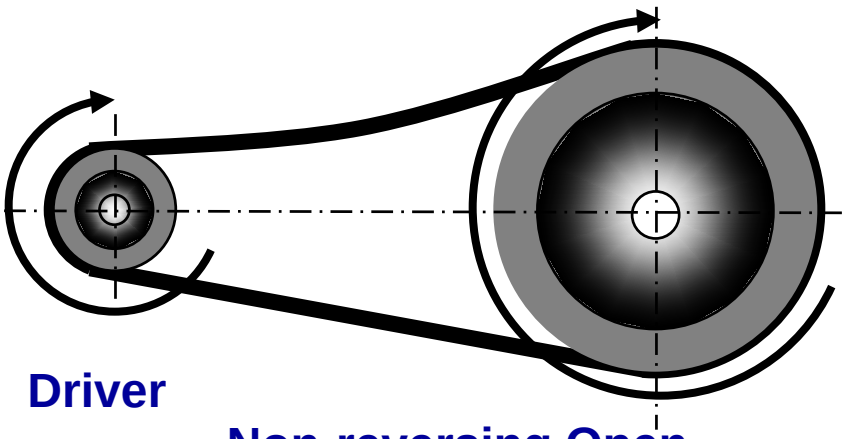
- Advantages

- Potential failure modes

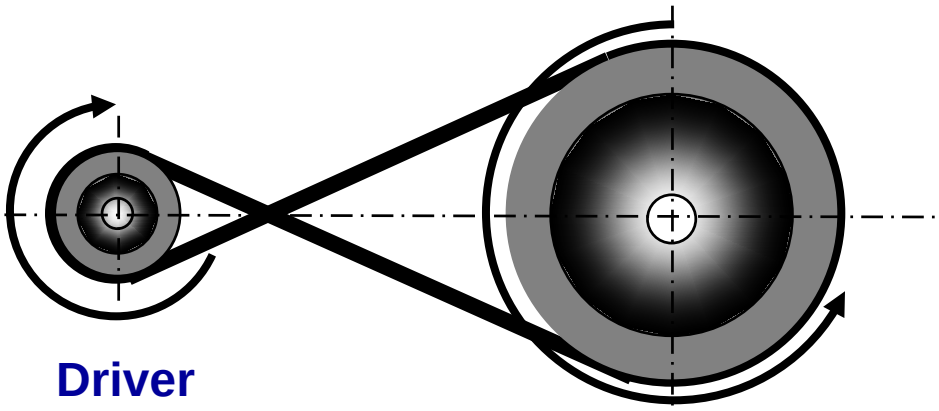
Belt Drive Overview
Team Exercise

- Advantages
  - Inexpensive - quiet
  - Span long distances - no lubrication (belts)
  - Absorb Shock Loads - light
  - Low Maintenance - no axial forces
  - Higher efficiencies
- Potential failure modes
  - Stretching - loss of friction (squealing belts)
  - Chain/Belt falls off the sprocket
  - Fatique Failures

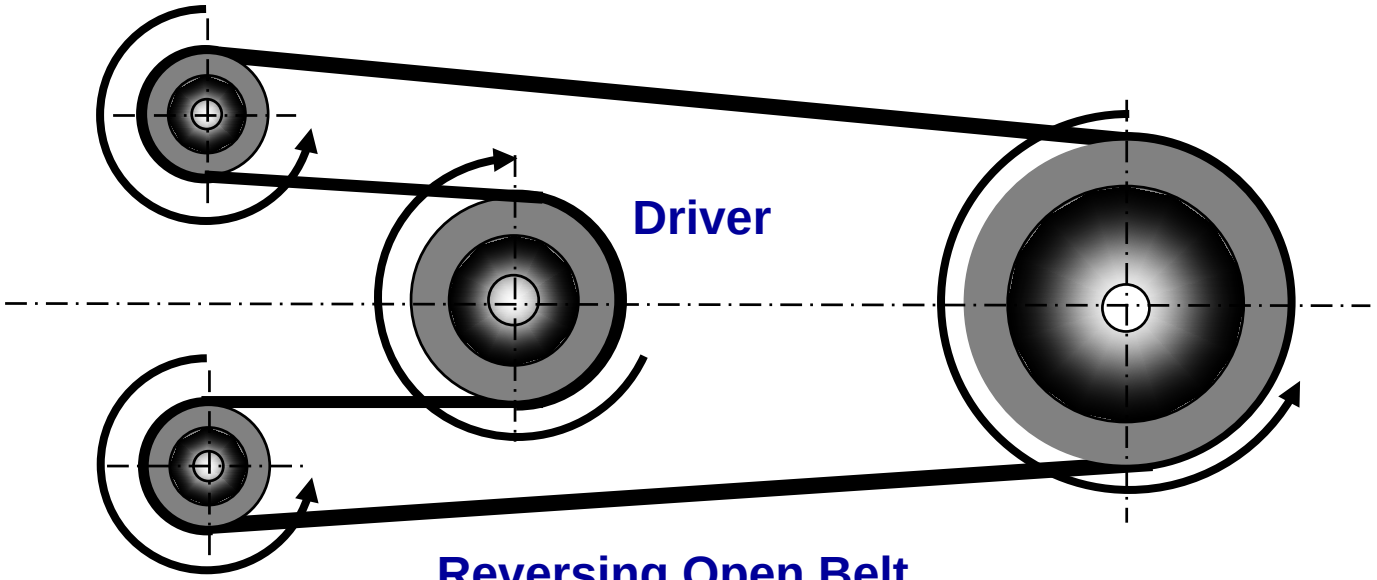
Flexible Drive Configurations



Non-reversing Open

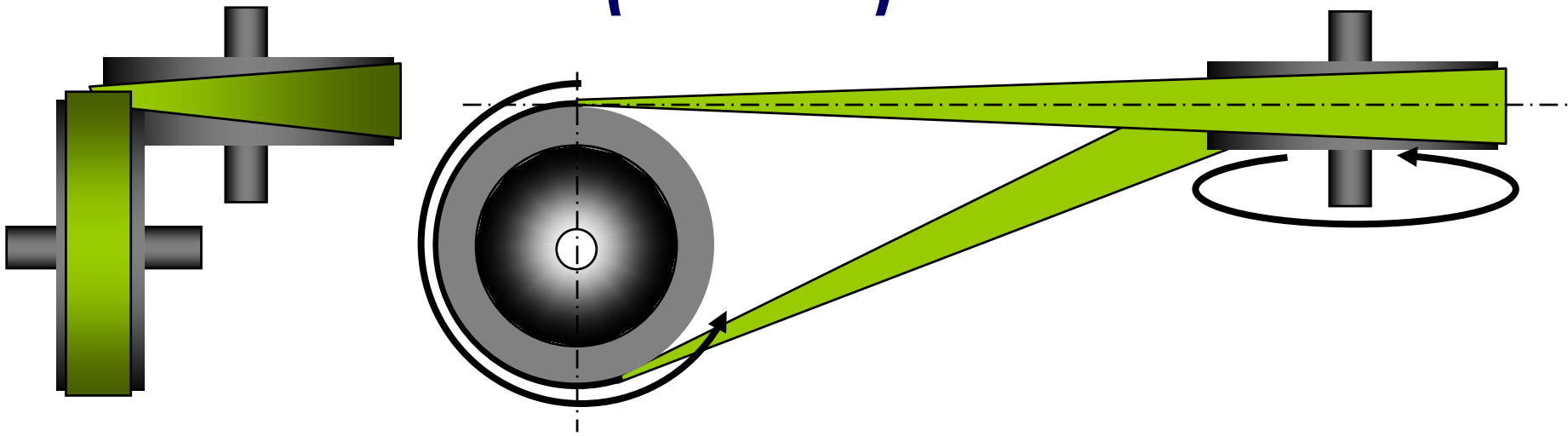


Reversing Cross Belt



Reversing Open Belt

Flexible Drive Configurations (cont'd)

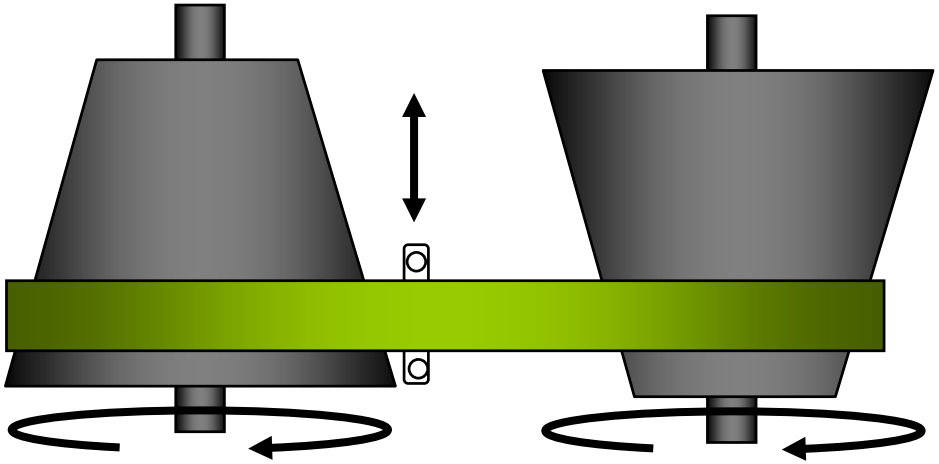


Twisted Belt Drive



Clutch-less disconnect

Flexible Drive Configurations (cont'd)



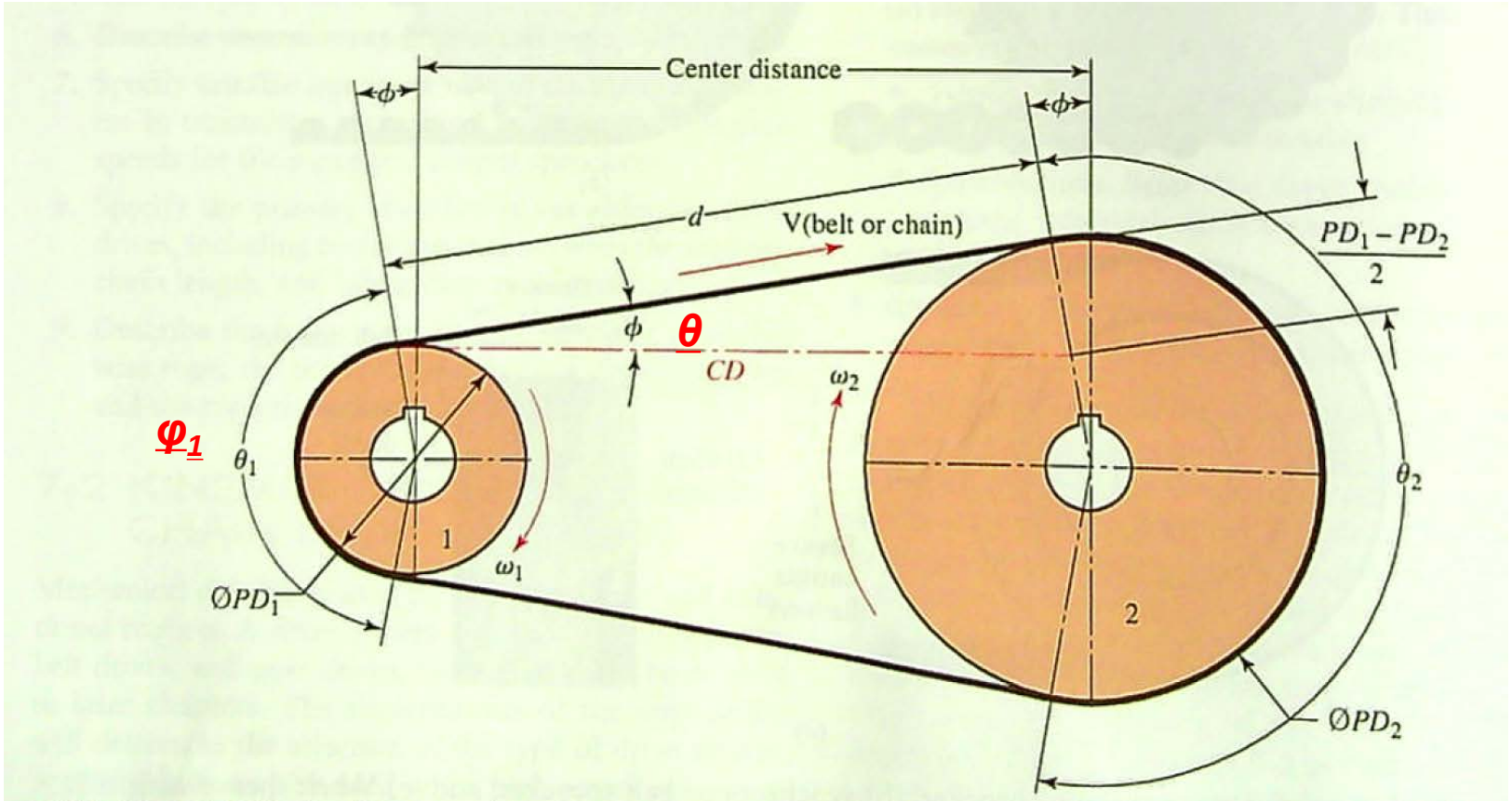
Continuously Variable



Continuous Variable Transmission



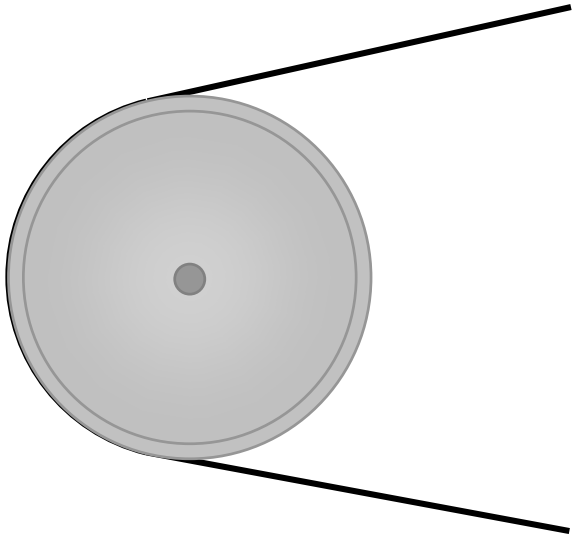
Belt Kinematics



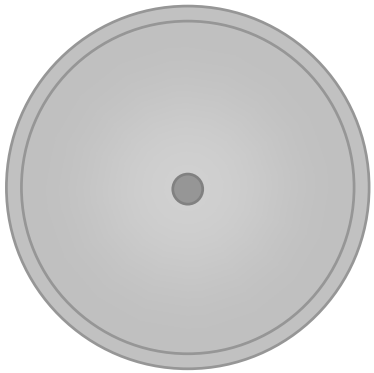
PD = Pitch Diameter Phi (theta) = Wrap Angle V = Belt(Chain) Speed Theta (phi) = Belt Angle

Belt Power Transmission

Transmission Parameters



Pulley FBD

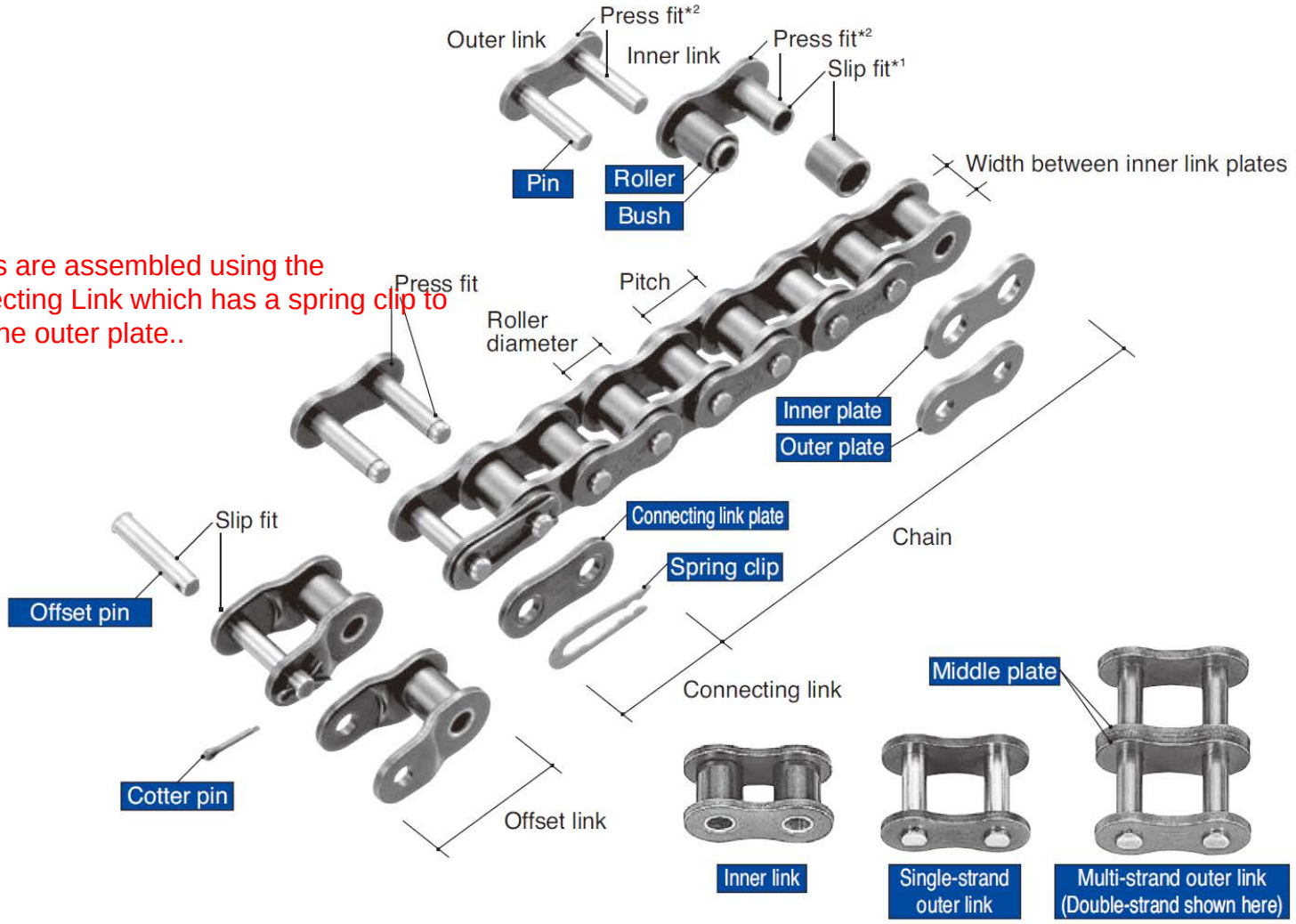


$R_x = (F_1 + F_2) \cos$

$R_y = (F_1 - F_2) \sin$

Roller Chain

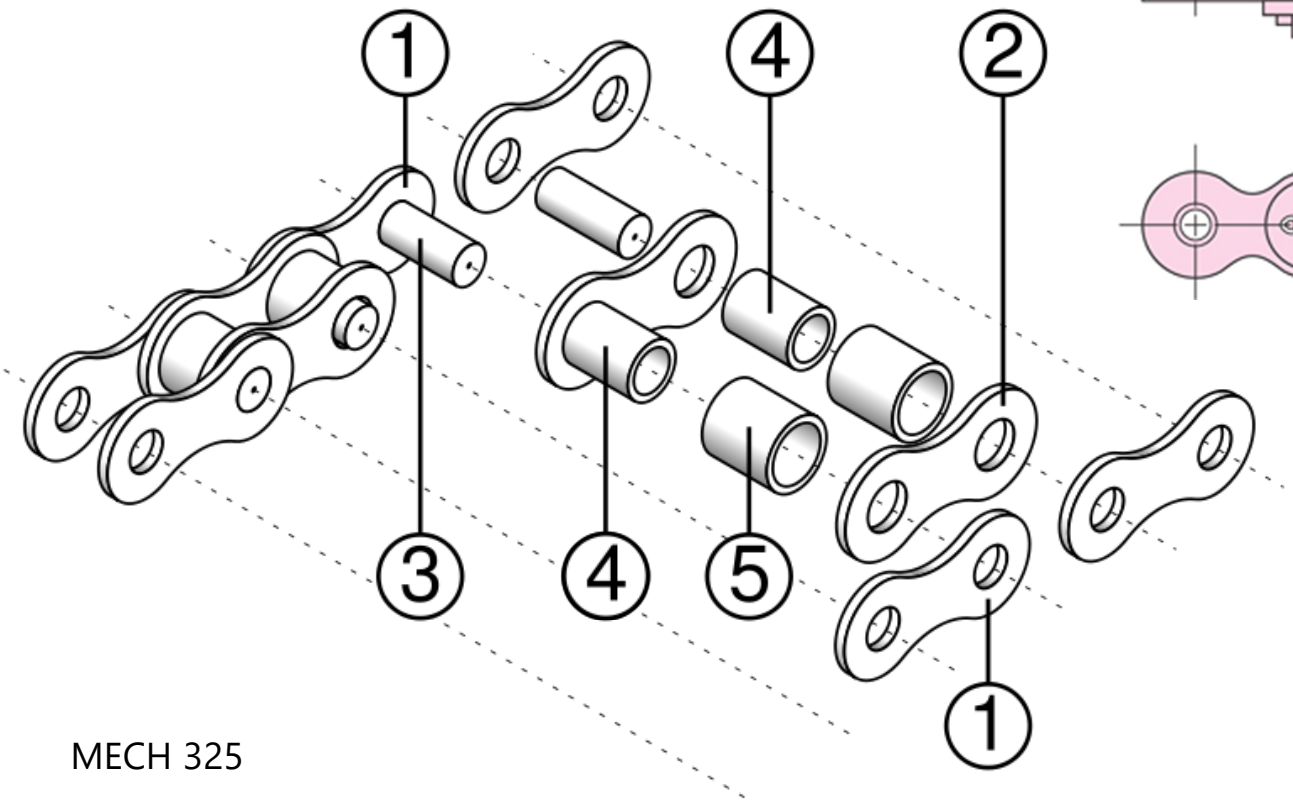
1. Basic Structure (Photo: RS Roller Chain)

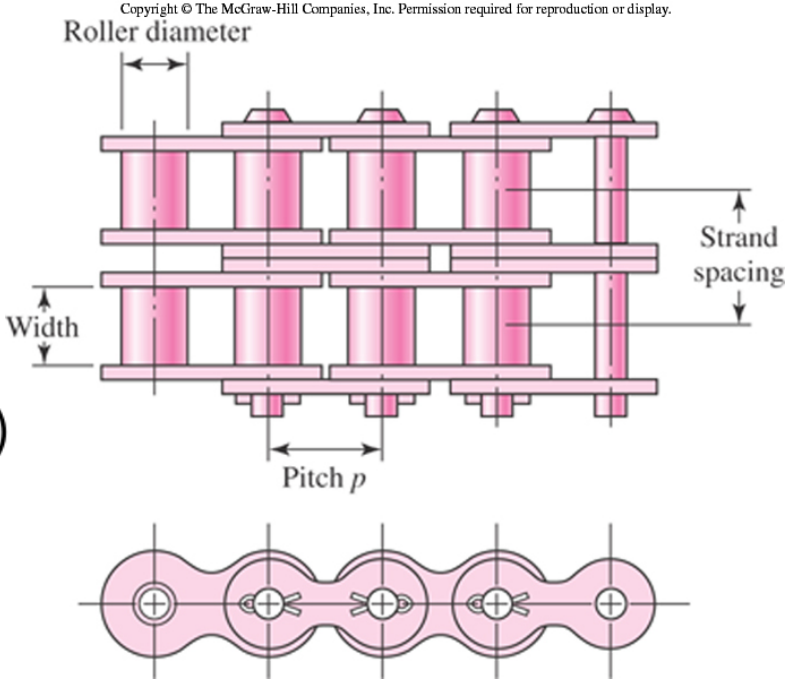


Chains are assembled using the Connecting Link which has a spring clip to Lock the outer plate..

Roller Chain

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1. Outer plate
2. Inner plate
3. Pin
4. Bushing
5. Roller

Chain Designation Sizes

TABLE 7–12 U.S. Roller Chain Sizes

Chain number	ISO 10823	Pitch (in)	Average tensile strength (lb)
25	4A	1/4	925
35	6A	3/8	2100
41		1/2	2000
40	8A	1/2	3700
50	10A	5/8	6100
60	12A	3/4	8500
80	18A	1	14 500
100	20A	$1\frac{1}{4}$	24 000
120	24A	$1\frac{1}{2}$	34 000
140	28A	$1\frac{3}{4}$	46 000
160	32A	2	58 000
180	36A	$2\frac{1}{4}$	80 000
200	40A	$2\frac{1}{2}$	95 000
240	48A	3	130 000

Reference: ANS1 Standard B29.1.

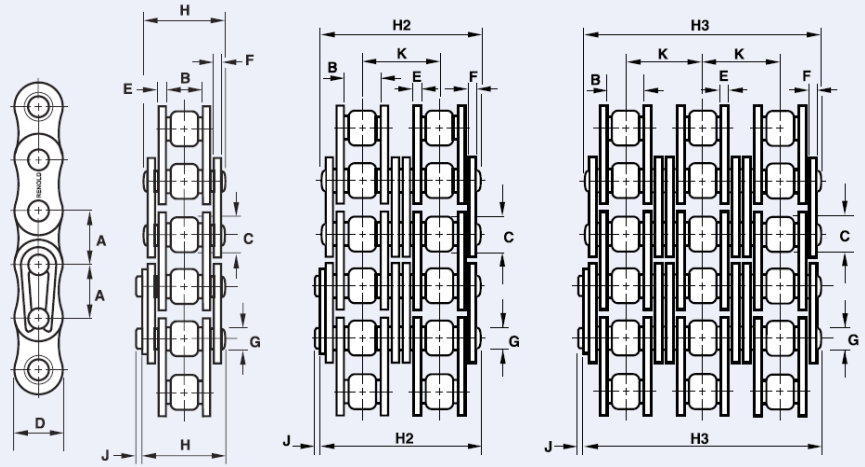
- Chains are designated by pitch sizes.
- The first digits are the number of 1/8 inches.
  - Example #40 chain has a pitch equal to 4 ×⅛″ = ½″
  - #100 chain = 10×⅛″ = 1¼″

Chain Selection

1 Standard

ANSI Standard Roller Chain

50 ANSI - American National Standards Institute

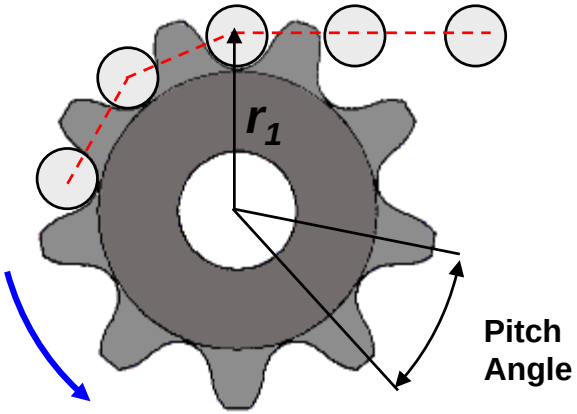


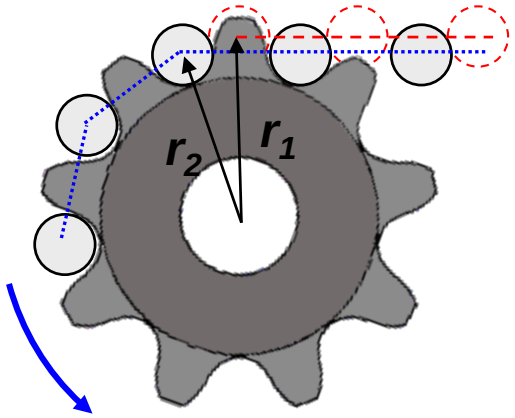
Note: Safety Factor

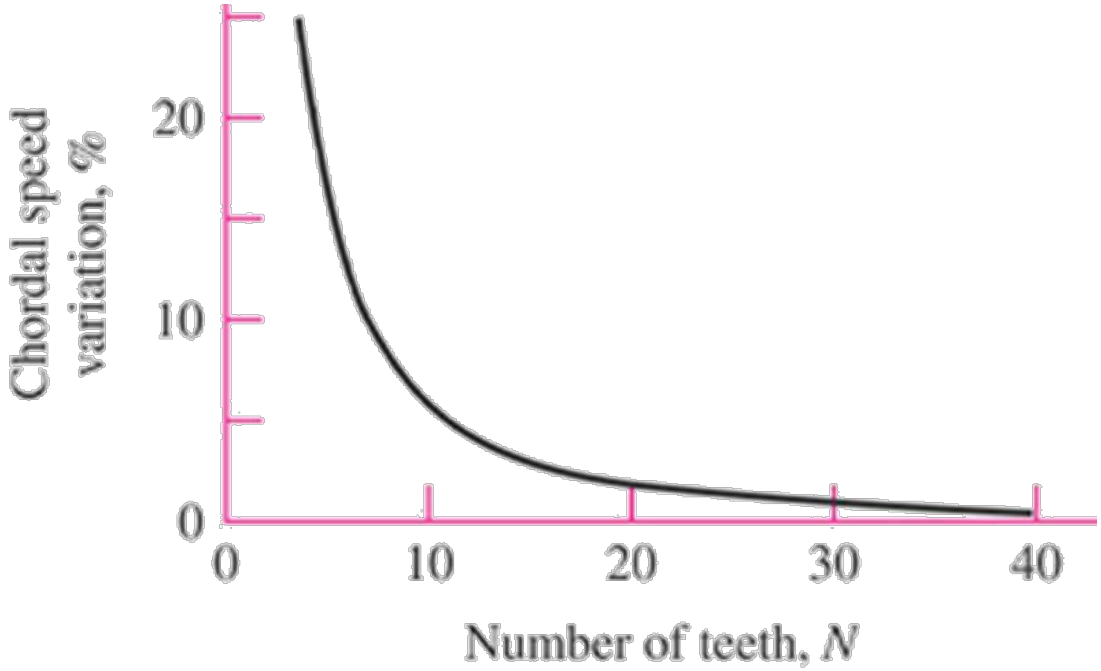
Dimensions are in inches unless otherwise indicated.

Chain No.	No. of Strands	Pitch	Inside Width Min	Roller Diam Max	Plate Height Max	Inner Plate Thick Max	Outer Plate Thick Max	Pin Diam Max	Pin Length Max	Conn Link Extra Max	Trans Pitch	Tensile Strength Min	Rated Working Load	Weight
		A	B	C	D	E	F	G	H	J	K	Lbs	Lbs	Lbs/Ft
50	1	0.625	0.370	0.400	0.594	0.080	0.080	0.200	0.803	0.043	—	4,880	1,400	0.67
50-2	2	0.625	0.370	0.400	0.594	0.080	0.080	0.200	1.512	0.043	0.713	9,760	2,380	1.41
50-3	3	0.625	0.370	0.400	0.594	0.080	0.080	0.200	2.224	0.043	0.713	14,640	3,500	2.01
50-4	4	0.625	0.370	0.400	0.594	0.080	0.080	0.200	2.963	0.043	0.713	19,520	4,620	2.68
50-5	5	0.625	0.370	0.400	0.594	0.080	0.080	0.200	3.672	0.043	0.713	24,400	5,460	3.35
50-6	6	0.625	0.370	0.400	0.594	0.080	0.080	0.200	4.385	0.043	0.713	29,280	6,440	4.02

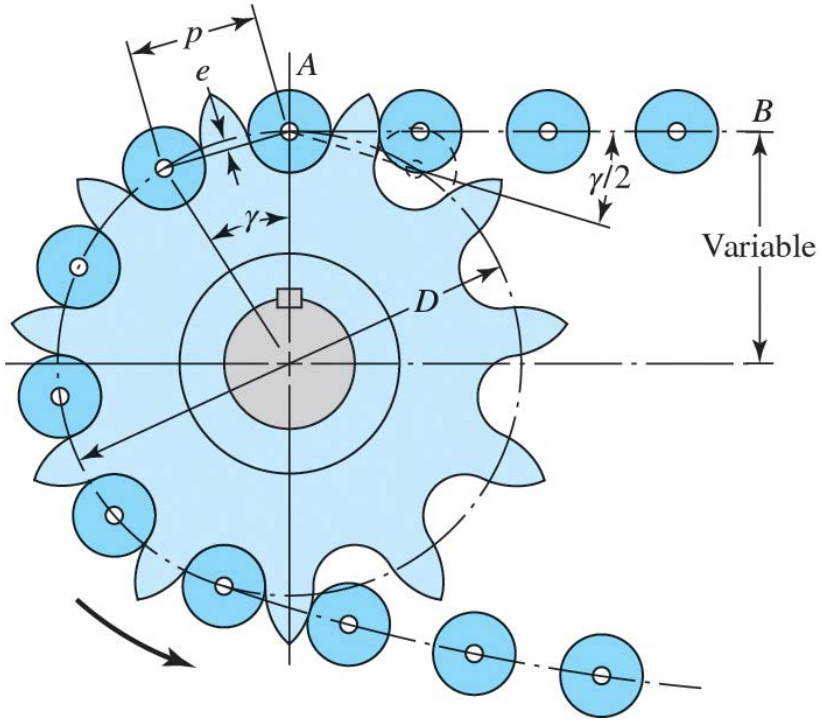
Chordal Speed Variation







Chain Equations



Pitch Diameter

$$D = \frac{p}{\sin(180°/N)}$$

Chain Velocity

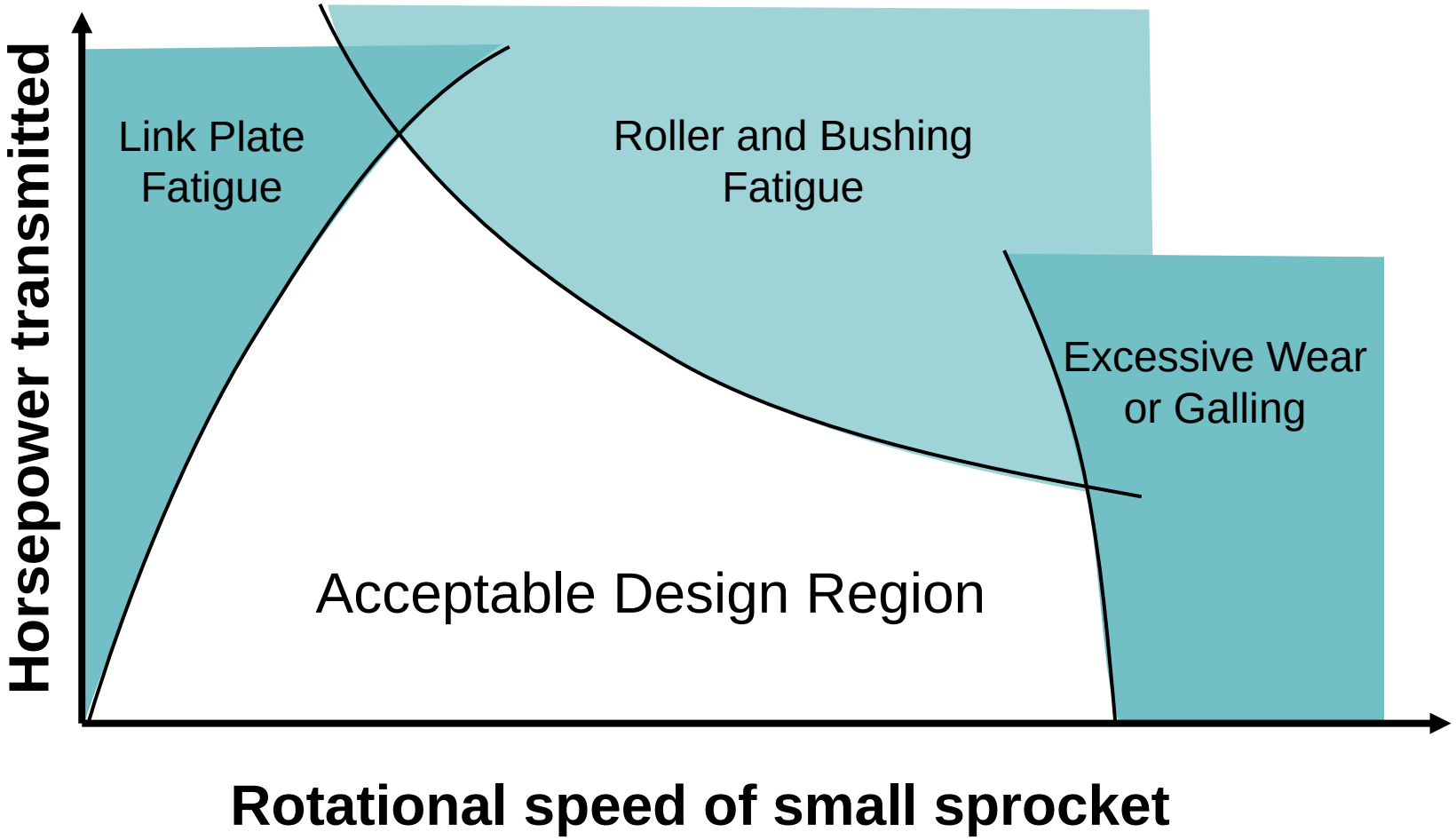
$$V = \frac{Npn}{12} \text{ feet per minute}$$

where N = number of sprocket teeth

p = chain pitch, in

n = sprocket speed, rev/min

Power Limits for Roller Chain



Power Limits for Roller Chain

- Link-plate failure (fatigue)

$$H_{\lim,lp} = K_{lp} N_1^{1.08} n_1^{0.9} p^{3.0-0.7p} \qquad K_{lp} = \begin{cases} 0.0022 & \text{no. 41 chain} \\ 0.004 & \text{other chains} \end{cases}$$

- Roller/bushing failure (fatigue)

$$H_{\lim,roller} = 1000 K_r N_1^{1.5} p^{0.8} n_1^{-1.5} \qquad K_r = \begin{cases} 28 & \text{no. 25, 35} \\ 3.4 & \text{no. 41} \\ 17 & \text{no. 40-240} \end{cases}$$

- Excessive wear / galling

$$H_{\lim,wear} = \left(\frac{n_1 N_1 p}{110.84}\right)(4.413 - 2.073p - 0.0274N_2) - \left(\ln\frac{n_2}{1000}\right)(1.59 \log p + 1.873)$$

N_1 = number of teeth of small sprocket N_2 = no. teeth of large sprocket
n_1 = rpm of small sprocket n_2 = rpm of large sprocket
p = pitch (inches)

Chain Selection (Mott)

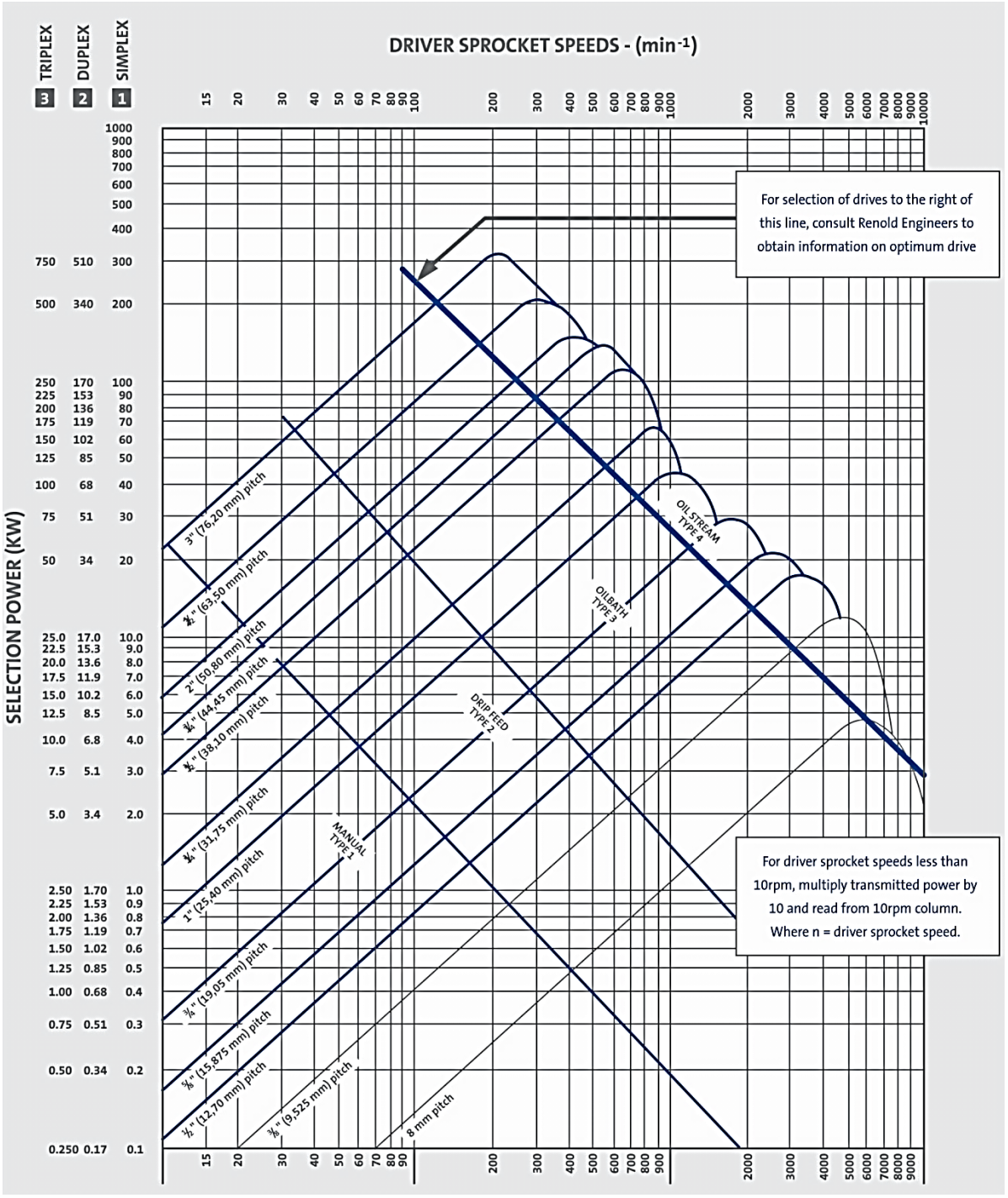
50 Horsepower Ratings – Single-Strand Roller Chain No. 50

No. of Teeth	Maximum Speed-Small Sprocket (rpm)																							
	10	25	50	100	140	200	300	500	700	900	1200	1500	1800	2100	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500
11	0.11	0.27	0.52	1.00	1.39	1.95	2.88	4.70	6.50	8.27	10.24	7.33	5.58	4.42	3.41	2.59	2.06	1.68	1.41	1.20	1.04	0.92	0.81	0.73	
12	0.12	0.29	0.56	1.09	1.51	2.13	3.14	5.13	7.09	9.02	11.67	8.35	6.35	5.04	3.88	2.95	2.34	1.92	1.61	1.37	1.19	1.04	0.93		
13	0.13	0.31	0.61	1.19	1.64	2.31	3.40	5.56	7.68	9.77	12.88	9.42	7.16	5.69	4.38	3.33	2.64	2.16	1.81	1.55	1.34	1.18			
14	0.14	0.34	0.66	1.28	1.76	2.48	3.67	5.99	8.27	10.53	13.87	10.52	8.01	6.35	4.89	3.72	2.95	2.42	2.03	1.73	1.50	0.28			
15	0.15	0.36	0.70	1.37	1.89	2.66	3.93	6.41	8.86	11.28	14.86	11.67	8.88	7.05	5.42	4.13	3.27	2.68	2.25	1.92	1.66				
16	0.16	0.39	0.75	1.46	2.02	2.84	4.19	6.84	9.45	12.03	15.85	12.86	9.78	7.76	5.98	4.55	3.61	2.95	2.47	2.11					
17	0.17	0.41	0.80	1.55	2.14	3.02	4.45	7.27	10.04	12.78	16.85	14.08	10.71	8.50	6.55	4.98	3.95	3.23	2.71	2.31					
18	0.18	0.43	0.84	1.64	2.27	3.19	4.71	7.70	10.63	13.53	17.84	15.34	11.67	9.26	7.13	5.42	4.30	3.52	2.95	0.05					
19	0.19	0.46	0.89	1.73	2.39	3.37	4.98	8.12	11.22	14.28	18.83	16.64	12.66	10.05	7.73	5.88	4.67	3.82	3.20						
20	0.20	0.48	0.94	1.82	2.52	3.55	5.24	8.55	11.81	15.04	19.82	17.97	13.67	10.85	8.35	6.35	5.04	4.13	3.46						
21	0.21	0.51	0.98	1.92	2.65	3.73	5.50	8.98	12.40	15.79	20.81	19.34	14.71	11.67	8.99	6.84	5.42	4.44							
22	0.22	0.53	1.03	2.01	2.77	3.90	5.76	9.41	12.99	16.54	21.80	20.73	15.77	12.52	9.64	7.33	5.82	4.76							
23	0.23	0.55	1.08	2.10	2.90	4.08	6.02	9.83	13.58	17.29	22.79	22.16	16.86	13.38	10.30	7.84	6.22	5.09							
24	0.24	0.58	1.13	2.19	3.02	4.26	6.28	10.26	14.18	18.04	23.78	23.62	17.97	14.26	10.98	8.35	6.63	1.36							
25	0.25	0.60	1.17	2.28	3.15	4.44	6.55	10.69	14.77	18.79	24.77	25.11	19.11	15.16	11.67	8.88	7.05								
26	0.26	0.63	1.22	2.37	3.28	4.61	6.81	11.12	15.36	19.55	25.76	26.64	20.26	16.08	12.38	9.42	7.47								
28	0.28	0.67	1.31	2.55	3.53	4.97	7.33	11.97	16.54	21.05	27.75	29.77	22.65	17.97	13.84	10.52	4.74								
30	0.30	0.72	1.41	2.74	3.78	5.32	7.86	12.83	17.72	22.55	29.73	33.01	25.11	19.93	15.34	11.67									
32	0.32	0.77	1.50	2.92	4.03	5.68	8.38	13.68	18.90	24.06	31.71	36.37	27.67	21.96	16.90	12.86									
35	0.35	0.84	1.64	3.19	4.41	6.21	9.16	14.97	20.67	26.31	34.68	41.60	31.65	25.11	19.34	0.94									
40	0.40	0.96	1.88	3.65	5.04	7.10	10.47	17.10	23.63	30.07	39.64	49.11	38.67	30.68	23.62										
45	0.45	1.08	2.11	4.10	5.67	7.98	11.78	19.24	26.58	33.83	44.59	55.24	46.14	36.61	8.64										
	Type A				Type B							Type C													

Type A: Manual or Drip Lubrication • Type B: Bath or Disc Lubrication • Type C: Oil Stream Lubrication

Multiple-Strand Factors					
Number of Strands	2	3	4	5	6
Factor	1.7	2.5	3.3	3.9	4.6

Ref: Roller Chain Power Limits (with Lubrication)



Source: Renold Chain
www.renoldcanada.com

Chain Lubrication

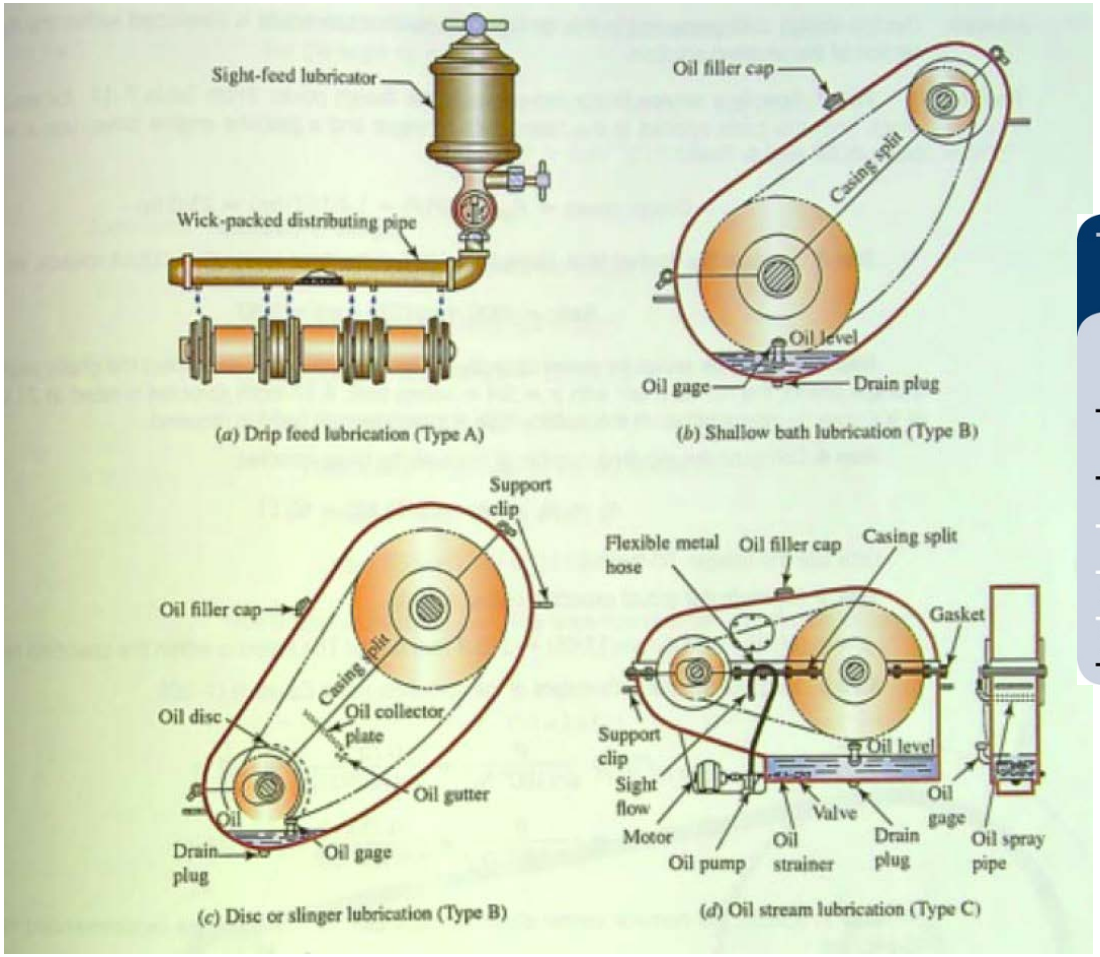


TABLE 7–18 Recommended Lubricant for Chain Drives

Ambient temperature		Recommended lubricant
°F	°C	
20 to 40	−7 to 5	SAE 20
40 to 100	5 to 38	SAE 30
100 to 120	38 to 49	SAE 40
120 to 140	49 to 60	SAE 50

Chain Selection

- Nominal power is defined as the expected rated (working) power for the machine.

- Apply a service factor, K, to the nominal power based on the expected use of the chain.
- The allowable power is the rated power times the service factor

TABLE 7–17 Service Factors for Chain Drives

Load type	Type of driver		
	Hydraulic drive	Electric motor or turbine	Internal combustion engine with mechanical drive
Smooth Agitators; fans; generators; grinders; centrifugal pumps; rotary screens; light, uniformly loaded conveyors	1.0	1.0	1.2
Moderate shock Bucket elevators; machine tools; cranes; heavy conveyors; food mixers and grinders; ball mills; reciprocating pumps; woodworking machinery	1.2	1.3	1.4
Heavy shock Punch presses; hammer mills; boat propellers; crushers; reciprocating conveyors; rolling mills; logging hoists; dredges; printing presses	1.4	1.5	1.7

Chain Selection

- Capacity (Tabulated) Tables based on the following parameters
  - 15000 hours full load (~21 months @24/7 operation)
  - 17 teeth sprocket
  - Speed ratio 1:1
  - Single Strand of 100 links
  - Service Factor = 1.0
  - Recommended Lubrication
  - **Minimum of Pre & Post-Extreme**
- Guidelines
  - Centerline distance, CD, should not be less than 150% of larger sprocket diameter and CD (in pitches) should be between 30 and 60
  - Adjust CD to actual chain size (in pitches or links)
  - Try to select sprocket with odd number of teeth and chain with even number of links.

50 Horsepower Ratings – Single-Strand Roller Chain No. 50

No. of Teeth	Maximum Speed-Small Sprocket													
	10	25	50	100	140	200	300	500	700	900	1200	1500	1800	210
11	0.11	0.27	0.52	1.00	1.39	1.95	2.88	4.70	6.50	8.27	10.24	7.33	5.58	4.4
12	0.12	0.29	0.56	1.09	1.51	2.13	3.14	5.13	7.09	9.02	11.67	8.35	6.35	5.0
13	0.13	0.31	0.61	1.19	1.64	2.31	3.40	5.56	7.68	9.77	12.88	9.42	7.16	5.6
14	0.14	0.34	0.66	1.28	1.76	2.48	3.67	5.99	8.27	10.53	13.87	10.52	8.01	6.3
15	0.15	0.36	0.70	1.37	1.89	2.66	3.93	6.41	8.86	11.28	14.86	11.67	8.88	7.0
16	0.16	0.39	0.75	1.46	2.02	2.84	4.19	6.84	9.45	12.03	15.85	12.86	9.78	7.7
17	0.17	0.41	0.80	1.55	2.14	3.02	4.45	7.27	10.04	12.78	16.85	14.08	10.71	8.5
18	0.18	0.43	0.84	1.64	2.27	3.19	4.71	7.70	10.63	13.53	17.84	15.34	11.67	9.2
19	0.19	0.46	0.89	1.73	2.39	3.37	4.98	8.12	11.22	14.28	18.83	16.64	12.66	10.0
20	0.20	0.48	0.94	1.82	2.52	3.55	5.24	8.55	11.81	15.04	19.82	17.97	13.67	10.8
21	0.21	0.51	0.98	1.92	2.65	3.73	5.50	8.98	12.40	15.79	20.81	19.34	14.71	11.6
22	0.22	0.53	1.03	2.01	2.77	3.90	5.76	9.41	12.99	16.54	21.80	20.73	15.77	12.5
23	0.23	0.55	1.08	2.10	2.90	4.08	6.02	9.83	13.58	17.29	22.79	22.16	16.86	13.3
24	0.24	0.58	1.13	2.19	3.02	4.26	6.28	10.26	14.18	18.04	23.78	23.62	17.97	14.2
25	0.25	0.60	1.17	2.28	3.15	4.44	6.55	10.69	14.77	18.79	24.77	25.11	19.11	15.1
26	0.26	0.63	1.22	2.37	3.28	4.61	6.81	11.12	15.36	19.55	25.76	26.64	20.26	16.0
28	0.28	0.67	1.31	2.55	3.53	4.97	7.33	11.97	16.54	21.05	27.75	29.77	22.65	17.9
30	0.30	0.72	1.41	2.74	3.78	5.32	7.86	12.83	17.72	22.55	29.73	33.01	25.11	19.9
32	0.32	0.77	1.50	2.92	4.03	5.68	8.38	13.68	18.90	24.06	31.71	36.37	27.67	21.9
35	0.35	0.84	1.64	3.19	4.41	6.21	9.16	14.97	20.67	26.31	34.68	41.60	31.65	25.1
40	0.40	0.96	1.88	3.65	5.04	7.10	10.47	17.10	23.63	30.07	39.64	49.11	38.67	30.0
45	0.45	1.08	2.11	4.10	5.67	7.98	11.78	19.24	26.58	33.83	44.59	55.24	46.14	36.0
	Type A				Type B									

Chain Selection (Shigley)

- Determine allowable power, H_a based on modified tabulated power H_{tab}.

$$H_a = K_1 K_2 H_{tab}$$

Table 17–20

Rated Horsepower Capacity of Single-Strand Single-Pitch Roller Chain for a 17-Tooth Sprocket

Source: Compiled from ANSI B29.1-1975 information only section, and from B29.9-1958.

Sprocket Speed, rev/min	ANSI Chain Number 25	35	40	41	50	60
50	0.05	0.16	0.37	0.20	0.72	1.24
100	0.09	0.29	0.69	0.38	1.34	2.31
150	0.13*	0.41*	0.99*	0.55*	1.92*	3.32
200	0.16*	0.54*	1.29	0.71	2.50	4.30
300	0.23	0.78	1.85	1.02	3.61	6.20
400	0.30*	1.01*	2.40	1.32	4.67	8.03
500	0.37	1.24	2.93	1.61	5.71	9.81
600	0.44*	1.46*	3.45*	1.90*	6.72*	11.6
700	0.50	1.68	3.97	2.18	7.73	13.3
800	0.56*	1.89*	4.48*	2.46*	8.71*	15.0
900	0.62	2.10	4.98	2.74	9.69	16.7
1000	0.68*	2.31*	5.48	3.01	10.7	18.3
1200	0.81	2.73	6.45	3.29	12.6	21.6
1400	0.93*	3.13*	7.41	2.61	14.4	18.1
1600	1.05*	3.53*	8.36	2.14	12.8	14.8
1800	1.16	3.93	8.96	1.79	10.7	12.4
2000	1.27*	4.32*	7.72*	1.52*	9.23*	10.6
2500	1.56	5.28	5.51*	1.10*	6.58*	7.57
3000	1.84	5.64	4.17	0.83	4.98	5.76
	Type A		Type B			Type C

*Estimated from ANSI tables by linear interpolation.

Note: Type A—manual or drip lubrication; type B—bath or disk lubrication; type C—oil-stream lubrication.

(*Continued*)

Table 17–22

Tooth Correction Factors, K_1

Number of Teeth on Driving Sprocket	K_1 Pre-extreme Horsepower	K_1 Post-extreme Horsepower
11	0.62	0.52
12	0.69	0.59
13	0.75	0.67
14	0.81	0.75
15	0.87	0.83
16	0.94	0.91
17	1.00	1.00
18	1.06	1.09
19	1.13	1.18
20	1.19	1.28
N	$(N_1/17)^{1.08}$	$(N_1/17)^{1.5}$

Table 17–23

Multiple-Strand Factors, K_2

Number of Strands	K_2
1	1.0
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6
8	6.0

Chain Drive Design (Shigley)

$H_a = K_1 K_2 H_{tab}$

$H_{TAB} = P_{TAB}$ Design Power (Catalogue)

H_a Allowable Power

K_1 Tooth correction (Table 17-22)

K_2 Multiple Strand Factor (Table 17-23)

$H_d = H_{nom} K_s n_d$

H_{nom} = Nominal Power (motor rating)

H_d = Design Power

K_s service factor

n_d design factor (normally = 1.0)

$N_b > H_d / H_a$

N_b = required number of strands

Safety Factor $n_{fs} = N_b H_a / H_d$

Ref: Roller Chain Selection Procedure

1. Determine design power requirement from nominal power. Remember to include service factor
2. Tentatively select chain pitch
3. Tentatively select number of teeth on small sprocket (consider chordal speed variation). In general, avoid less than 17 teeth.
4. Determine required number of teeth on large sprocket (7:1 ratio is near maximum)
5. Check that design power is below limiting power
6. Check velocity V (should not exceed 9000 ft/min)
7. Determine chain length (integer number of pitches)

Chain Length

- Chains must be specified as an integer number of links
  - L/p = number of links
  - L = Chain length (inch)
  - p = pitch (inch)
- Normally an initial value for center distance, C, is selected
  - C should be between 30 and 50 links
- Determine L/p (round to integer link) and re-calculate C

Chain length in pitches

$$\frac{L}{p} \approx \frac{\sim 2C}{p} + \frac{N_1 + N_2}{2} + \frac{(N_2 + N_1)^2}{4\pi^2 C/p} \qquad \textbf{(17-34)}$$

Center-to-center distance

$$C = \frac{P}{4}\left[-A + \sqrt{A^2 - 8\left(\frac{N_2 - N_1}{2\pi}\right)^2}\right] \qquad \textbf{(17-35)}$$

$$A = \frac{N_1 + N_2}{2} - \frac{L}{P} \qquad \textbf{(17-36)}$$

Sprocket Selection

Martin

Application Data and Selection Procedure

Step I

Service Classification — Table I

Uniform Load

Agitators, Liquid
Blowers, Centrifugal
Conveyors, Even Load
Elevators, Even Load
Fans, Centrifugal
Generators
Line Shafts, Even Load
Machines, Even Load, Non-reversing
Pumps, Centrifugal

Moderate Shock Load

Beaters
Compressors, Centrifugal
Conveyors, Uneven Load
Elevators, Uneven Load
Grinders, Pulp
Kilns and Dryers
Laundry - Washers and Tumblers
Line Shafts, Uneven Load
Machines, Pulsating Load, Non-reversing
Pumps, Reciprocating, Triplex
Screens, Rotary, Even Load
Woodworking Machinery

Heavy Shock Load

Brick Machines
Compressors Reciprocating
Crushers
Machines, Reversing or Impact Loads
Mills, Hammer, Rolling or Drawing
Presses
Pumps, Reciprocating, Simplex or Duplex

Step II

Service Factor — Table II

SERVICE CLASSIFICATION	TYPE OF INPUT POWER		
	Internal Combustion Engine with Hydraulic Drive	Electric Motor or Turbine	Internal Combustion Engine with Mechanical Drive
Uniform Load	1.0	1.0	1.2
Moderate Shock Load	1.2	1.3	1.4
Heavy Shock Load	1.4	1.5	1.7

Unfavorable Operating Conditions which may be present should be compensated for by adding .2 to the Service Factor for each unfavorable condition. Some of these conditions are listed below:

1. Multiple Shafts — add .2 for each additional shaft.
2. Excessive speed ratios — exceeding 7 to 1.
3. Heavy starting loads with frequent starts and stops.
4. Conditions of high temperatures, unusually abrasive conditions, or circumstances decreasing lubrication effectiveness or not allowing the use of recommended lubrication procedures.

Step III

Determination of Design Horsepower

Determine the design horsepower of the required drive using the following procedure.

1. Determine Service Classification — Table I. Unlisted equipment may be classified by its similarity to a listed type.
2. Using Service Classification and Frequency of Service, select the Service Factor — Table II. Increase the Service Factor by adding compensation for unfavorable operating conditions.
3. Multiply the normal operating horsepower of the drive by the Compensated Service Factor to obtain Service Horsepower.

Step IV

Drive Selection

Using Design Horsepower computed above, use Trial Selection Chart (Horsepower Tables) on page E184-E185, or enter tables of Horsepower Ratings shown on pages E186 thru E192. Select the smallest pitch chain which has the required horsepower rating for a pinion sprocket turning at the specified RPM. Check to be certain the selected sprocket has a listed maximum bore large enough to accommodate the specified shaft. The tables on pages E-158 thru E-159 gives maximum bores for the usual range of driving sprockets.

If the Design Horsepower at the required RPM is greater than the horsepower rating of the largest pitch chain which can operate at that speed, a multiple chain drive should be considered for the application.

Selection of drives to operate at speeds somewhat below the maximum rating will increase the life of the drive and quietness of operation.

Step V

Driving Sprocket

In selecting the driving sprocket 17 **teeth are recommended as a minimum** although 15 teeth are quite often used, and as low as 7 teeth can be cut. When the maximum bore of the 17 tooth sprocket will not accomodate the driving shaft, it is necessary to go to a sprocket with a greater number of teeth. Hardened teeth are recommended for sprockets with 25 teeth or less.

Application Data and Selection Procedure

Step VI

Driven Sprocket (Ratio)

The number of teeth selected for the driven sprocket depends upon the driving sprocket chosen and the desired speed of the driven shaft. When space limitations are a factor, the diameter of the driven sprocket sometimes determines the final selection of drive.

The recommended maximum speed ratio is 7 to 1, although higher ratios are occasionally used. It is usually better design, however, for large reductions to use a double reduction drive.

Select the driven sprocket size from the Speed Ratio Table on page E-170 using the required speed ratio and size of driver sprocket.

Step VII

Shaft Centers

May be calculated from the formula on page E-168 where the sprocket diameters and chain length are known.

On many applications the motor base is adjustable, allowing for slight changes in shaft centers. On long centers some form of chain adjuster or take-up is recommended.

Step VIII

Chain Length

On page E-168 is shown a simple method of computing the length of chain necessary for a drive with given sprocket dimensions and center to center distance of shafts. (See chart on page E-169 for length in ft.)

Chain Drive Design Example

To select a roller chain drive from a 10 HP electric motor (1⅝" shaft) 1200 RPM (1150 under load) to a wood working machine shaft at 300 RPM on 30" centers. Drive conditions — moderate pulsating load, good lubrication, 10 hour day operation.

1. Service class — moderate shock load (Table I).
2. Service factor — 1.3 (Table II).
3. Design HP — 1.3 × 10 = 13 DHP.
4. Selection — The Horsepower Ratings on page E-184 show that either of the following combinations may be used.

   No. D40-19 Tooth — Smoothest in operation

   No. 50-18 Tooth — Lower drive cost

For our purpose we select No. 50 chain and checking the bore find that the 1⅝" shaft can be accommodated with a stock bored to size sprocket.

The driven sprocket is found as follows:

No. Teeth
Driven
Sprocket = $18 \times \frac{1150}{300}$ (Ratio) = 68.99 or 69 Teeth

Since 69 teeth is not a stock size we select 70 teeth. The chain length is calculated as shown on page E-169 and is 142 pitches.

Overhung Load

When a Sprocket is mounted on a reducer shaft, a calculation should be made to determine the overhung load in pounds using formula on page *i*-2 in general engineering section.

Roller Chain Life

- Wear is the normal mode of failure
- Under normal conditions, chain properly maintained, life = 8×10^6 cycles or 15,000 hrs (whichever comes first)
- Life reduced up to 300 times when no lubrication used
- General rule: replace sprockets every third time chains are replaced
- Sprocket teeth are often hardened to reduce wear

Induction Hardening

https://www.youtube.com/watch?v=b8Ptfw39ea4



Ref: Roller Chain Life

- Recommended nominal safety factor **on breaking** for a single-strand roller chain shown below (yield safety factors ~150% higher)

Nominal Safety Factor	Cycles (x 10^6)	Application details
5.0*	1.0	Dynamic load does not exceed working load
6.0*	2.0	
8.0	8.0	Dynamic loads occasionally exceed working load by 20%
10.0	8.0	Passenger lifts
12.0	8.0	Harsh environments

*Note: safety factors below 8.0 result in increased wear rate unless special attention is given to improving lubrication

Source: Renolds Roller Chain (www.renolds.com)

Ref: Factors Affecting Chain Life

Recommended increases to **design factor, n_d** based on lubrication, cleanliness, temperature, and load type.

Lubrication	Cleanliness		
	Clean	Moderately Clean	Dirty / Abrasive
Regular	1.00	1.25	1.50
Occasional	1.25	1.50	1.75
None	1.50	1.50	1.75

Temperature (^{0}C)	Load Regime		
	Smooth	Moderately Shocks	Heavy Shocks
$+10^0$ to $+150^0$	1.00	1.34	1.88
0^0 to 10^0	1.25	1.88	2.38
-20^0 to 0^0	1.50	2.50	3.13
-40^0 to -20^0	1.88	3.13	4.13

Source: Renolds Roller Chain (www.renolds.com)